



PRODUCT DATA SHEET

ELECTRICAL MOTOR OIL – TECHNOFLUID SUBMERSIBLE MOTOR OIL

Commercial / Pack Name: SUBMERSIBLE MOTOR OIL

Product Description

- TECHNOFLUID Submersible Motor Oil is a high-quality insulating and lubricating oil specially formulated for use in water-filled and oil-filled submersible pump motors.
- Manufactured from highly refined base oils with excellent dielectric and thermal properties.
- Designed to provide effective lubrication of motor bearings while ensuring efficient heat dissipation.
- Exhibits excellent electrical insulating properties to protect motor windings.
- Maintains stable performance under continuous submerged operating conditions.

Key Performance Benefits

- High dielectric strength ensures reliable electrical insulation of motor windings.
- Excellent thermal conductivity aids in efficient heat transfer and cooling.
- Superior lubrication protects bearings and moving parts, extending motor life.
- Good oxidation stability minimizes oil degradation during prolonged operation.
- Compatible with copper windings, motor insulation materials, and seals.
- Helps reduce motor failure, maintenance frequency, and downtime.

Applications Recommended for use in:

- Submersible pump motors
- Borewell and open well submersible pumps
- Agricultural irrigation pumps
- Domestic and industrial water pumping systems

Suitable for both single-phase and three-phase submersible motors (as per OEM recommendation).

Typical Uses

- Electrical insulation and lubrication medium in submersible motor assemblies.

- Motors operating continuously under water with high thermal and electrical loads.
- Applications requiring long oil life and stable insulating performance.

Performance Standards Meets or exceeds requirements as per:

- IS 335 (where applicable)
- OEM specifications for submersible motor oils

Operating Characteristics

- Maintains stable viscosity over a wide temperature range.
- Low moisture content ensures consistent insulating performance.
- Non-corrosive and compatible with commonly used motor materials.

Typical Characteristics

Characteristics	Test Method	Typical Figures
Appearance	Visual	Clear & Bright
Colour	ASTM D1500	≤ 1.0
Base Oil Type	—	Highly Refined Mineral
Density @ 29.5°C, g/ml	ASTM D4052	0.82 – 0.845
Kinematic Viscosity @ 40°C, cSt	ASTM D445	20 – 30
Kinematic Viscosity @ 100°C, cSt	ASTM D445	4.0 – 5.5
Viscosity Index	ASTM D2270	≥ 95
Flash Point (COC), °C	ASTM D92	≥ 200
Pour Point, °C	ASTM D97	-6
Dielectric Strength (BDV), kV	IEC 60156 / IS 6792	≥ 40
Moisture Content, ppm	ASTM D6304	≤ 50
Copper Corrosion (3 hrs @ 100°C)	ASTM D130	1a
Oxidation Stability	ASTM D943	Excellent
Neutralization Value, mg KOH/g	ASTM D664	≤ 0.03
Foaming Characteristics	ASTM D892	Pass
Thermal Conductivity	—	Good
Compatibility with Motor Insulation	In-house	Excellent

